GAN

Structured controls

PC-AVS / A2V Priors
Fast: Compact motion controls
Stable: Control leakage and jitter

Regional synthesis

MuseTalk / FastTalk
Fast: Restrict the synthesized region
Stable: Crop seams and framewise flicker

Personalized synthesis

Live Speech Portraits / GANimate
Fast: Reuse enrolled appearance
Stable: Controller or latent-search variation

Codebooks and detail retrieval

FlowVQTalker
Fast: Predict compact visual codes
Stable: Quantization and detail consistency

Rendering

Parametric geometry and UV features

StyleAvatar / RAM-Avatar
Fast: Reuse geometry and UV features
Stable: Tracking and deformation error

Implicit appearance transport

LivePortrait / DINet
Fast: Warp persistent source features
Stable: Occlusion and identity leakage

Neural radiance fields

AD-NeRF / ER-NeRF
Fast: Accelerate field queries
Stable: Geometry and expression mismatch

Gaussian scenes

FlashAvatar / LAM
Fast: Rasterize compact primitives
Stable: Deformation error and floaters

Diffusion

Few-step and one-step sampling

FADA / TurboTalk
Fast: Reduce sampler evaluations
Stable: Temporal consistency after distillation

Streaming denoising

Live Avatar / SoulX-FlashTalk
Fast: Chunk execution and cache reuse
Stable: Chunk boundaries and history drift

Temporal state and recovery

MEMO / InfiniteTalk
Fast: Limit repeated context processing
Stable: Anchoring and recovery from drift

Visual AR

Joint video-motion tokens

UniMo
Fast: Share a compact token sequence
Stable: Alignment and error propagation

Causal visual state + local emission

Body of Her / MIDAS
Fast: Reuse history; emit frames locally
Stable: Cache policy and recurrent drift

Phoneme-guided keyframes

AvatarSync
Fast: Sparse keyframes plus interpolation
Stable: Interpolation flicker; causality limits